Calcurion Maths Worksheets

A-Level Mathematics

Topic: Statistics – Correlation & Regression

SOLUTIONS

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1. As one variable increases, the other tends to increase.
2. As one increases, the other tends to decrease.
3. $-1 \leq r \leq 1$.
4. Strong positive linear correlation.
5. Weak negative linear correlation.
6. No linear correlation.
7. For each unit increase in x, y increases by 2.5 on average.
8. $y = 20$.
9. Relationship may not hold outside observed range; unreliable.
10. $S_{xy} = 420 - \frac{50\times80}{10} = 20$; $S_{xx} = 300 - \frac{50^2}{10} = 50$.
11. $b = \frac{20}{50} = 0.4$.
12. $y - 8 = 1.2(x - 5) \Rightarrow y = 1.2x + 2$.
13. x (explanatory).
14. Strong positive correlation.
15. Lurking variables; coincidence; reverse causation.
16. $S_{xy} = 310 - \frac{40\times56}{8} = 30$; $S_{xx} = 30$; $S_{yy} = 28$; $r = \frac{30}{\sqrt{30\times28}} \approx 1.04$? Recheck arithmetic; $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$.
17. $b = \frac{S_{xy}}{S_{xx}}$; line through $(\bar{x}, \bar{y})$.
18. $\bar{y} = 0.8 \times 10 + 5 = 13$.
19. y on x minimises vertical errors; x on y minimises horizontal.
20. $S_{xy} = r\sqrt{S_{xx}S_{yy}} = 0.7\sqrt{4000}$.
21. Point poorly predicted by the line.
22. Outlier had large influence; correlation sensitive to extremes.
23. $r^2 = 0.64$; 64% of variation in y explained by linear relation with x.
24. $y - 20 = 1.5(x - 10) + 3 \Rightarrow y = 1.5x + 8$.
25. Points closely follow a downward sloping line; strong negative correlation.